# Assignment 4

## Question 1

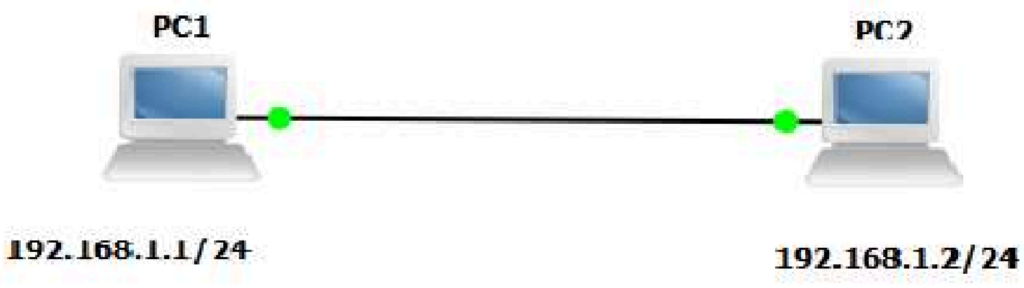
Discuss the GNS simulation environment.

### Solution

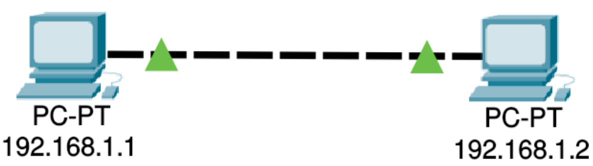
The GNS (G) simulation environment is a powerful tool designed to facilitate the simulation of various network protocols and systems. It provides a flexible framework that allows users to create, configure, and run simulations of network scenarios. The environment supports a wide range of protocols, including TCP/IP, UDP, and others, enabling users to model complex network behaviors.

## Question 2

Connect one PC with another PC as per diagram below.




### Solution

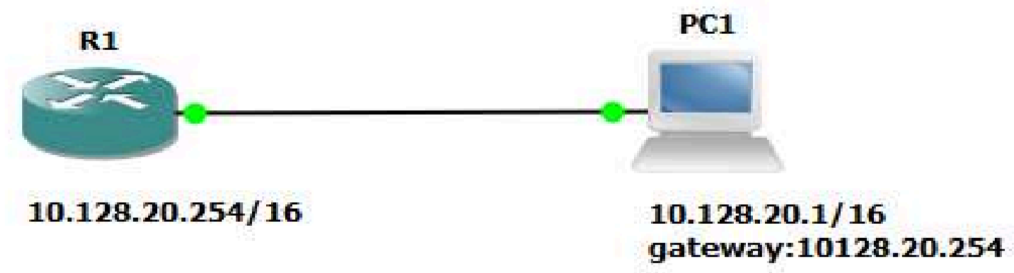



**Ping Results** (from `192.168.1.1`)

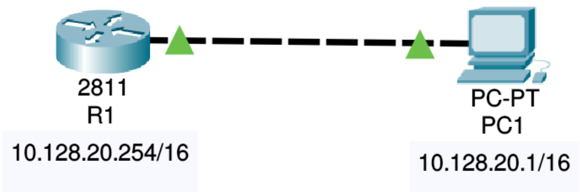
```txt
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
  Packets: Sent = 3, Received = 3, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
  Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

## Question 3

Connect a PC with Router Ethernet port as shown below.




## Solution




### Simulation Results

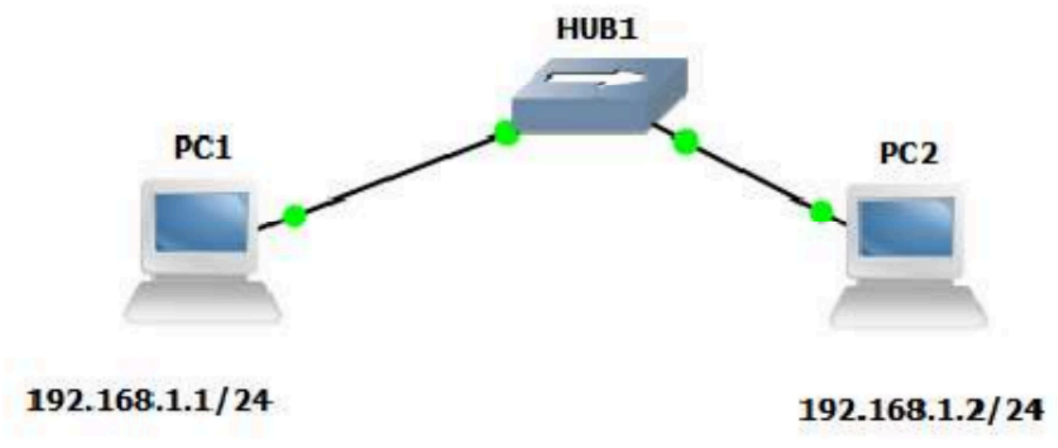
| Fire | Last Status | Source | Destination | Type | Color | Time(sec) | Periodic | Num | Edit | Del |
|---|---|---|---|---|---|---|---|---|---|---|
| | Successful | Router5 | 10.128.20.1 | ICMP | | 5.000 | N | 0 | (edit) | |

### Ping Results (from `10.128.20.1`)

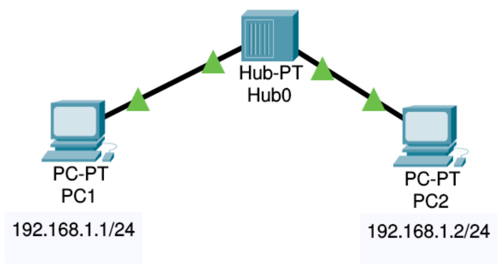
```txt
C:\>ping 10.128.20.254

Pinging 10.128.20.254 with 32 bytes of data:

Reply from 10.128.20.254: bytes=32 time<1ms TTL=255
Reply from 10.128.20.254: bytes=32 time<1ms TTL=255
Reply from 10.128.20.254: bytes=32 time<1ms TTL=255
Reply from 10.128.20.254: bytes=32 time<1ms TTL=255

Ping statistics for 10.128.20.254:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

## Question 4

Create a network of 2 PCs as shown below.




## Solution

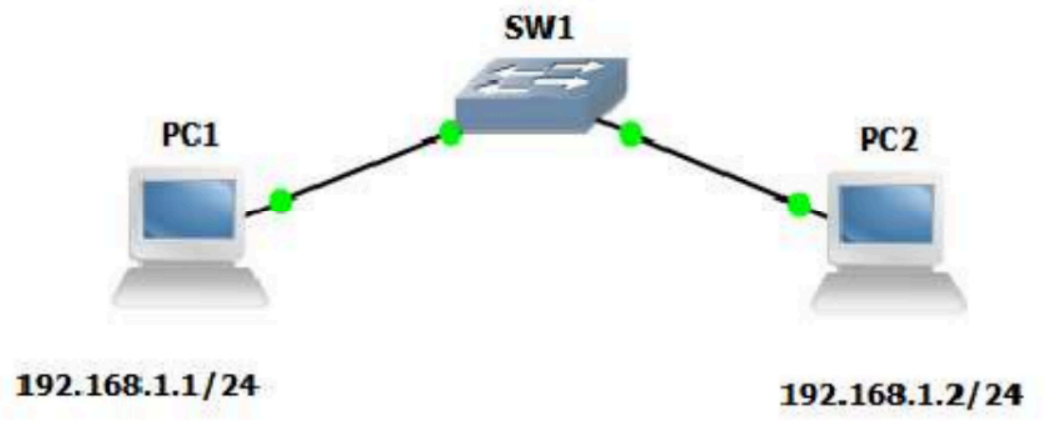



**Ping Results** (from 192.168.1.1)

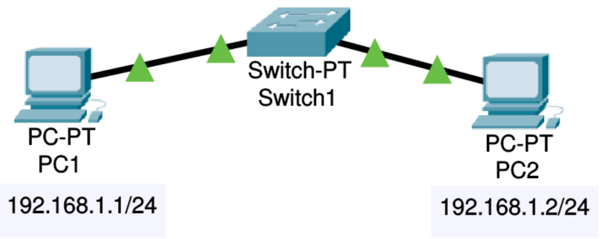
```txt
C:>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

## Question 5

Connect 2 PCs with a Switch below.




## Solution




### Simulation Results

| Fire | Last Status | Source | Destination | Type | Color | Time(sec) | Periodic | Num | Edit | Delete |
|---|---|---|---|---|---|---|---|---|---|---|
| | Successful | PC1 | 192.168.1.2 | IC... | | 1.000 | N | 0 | (e... | |

### **Ping Results** (from `192.168.1.1`)

```txt
Query successful

C:>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Reply from 192.168.1.2: bytes=32 time=1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time<1ms TTL=128
Reply from 192.168.1.2: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.2:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
```